Quadratics, Exponentials & Sequences

A Graphing and Skills Practice Worksheet

Level: Algebra 1

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Part A · Graphing Quadratic Functions

Every quadratic function can be written in three equivalent forms. Each form makes different features easy to read directly off the equation.

Form	Equation	What you can read directly	Direction / Width
Standard Form	$y = ax^2 + bx + c$	y-intercept $= (0, c)$; axis of symmetry $x = -\dfrac{b}{2a}$; vertex $= \left(-\dfrac{b}{2a}, f\left(-\dfrac{b}{2a}\right)\right)$	$a > 0$: opens up $a < 0$: opens down $\lvert a\rvert > 1$: narrower $\lvert a\rvert < 1$: wider
Vertex Form	$y = a(x - h)^2 + k$	Vertex $= (h, k)$ directly; axis of symmetry $x = h$	
Intercept (Factored) Form	$y = a(x - p)(x - q)$	x-intercepts $= (p, 0)$ and $(q, 0)$ directly; axis of symmetry $x = \dfrac{p + q}{2}$	

Note: in every form, a controls whether the parabola opens up or down and how narrow or wide it is.

A.1 Standard Form: $y = ax^2 + bx + c$

Worked Example

Graph $y = x^2 - 4x + 3$.

- y-intercept: $c = 3 \Rightarrow (0, 3)$
- Axis of symmetry: $x = -\dfrac{b}{2a} = -\dfrac{-4}{2(1)} = 2$
- Vertex: $x = 2 \Rightarrow y = 4 - 8 + 3 = -1 \Rightarrow (2, -1)$
- Factor to find x-intercepts: $x^2 - 4x + 3 = (x - 1)(x - 3) = 0 \Rightarrow x = 1, 3$
- $a = 1 > 0 \Rightarrow$ opens up



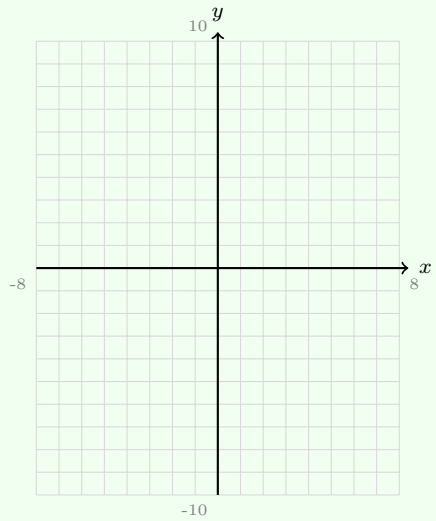
Easy

(a) Graph $y = x^2 - 2x - 3$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down

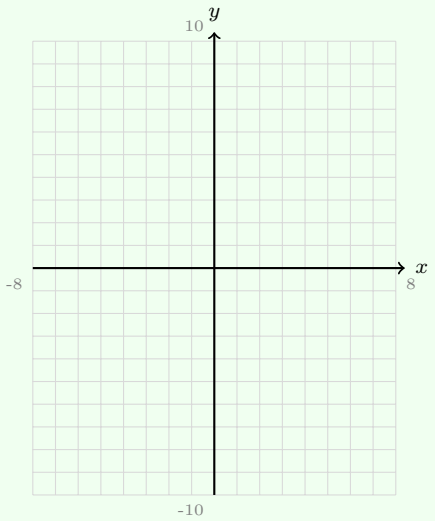


(b) Graph $y = x^2 + 4x$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down



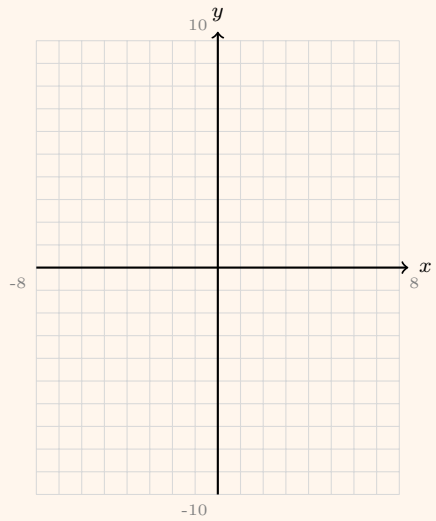
Medium

(a) Graph $y = x^2 - 6x + 5$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down

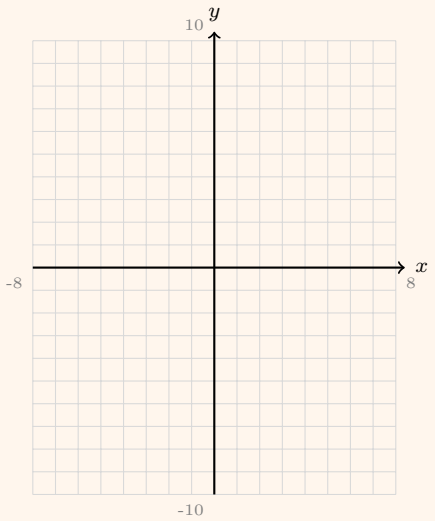


(b) Graph $y = 2x^2 + 4x - 6$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down



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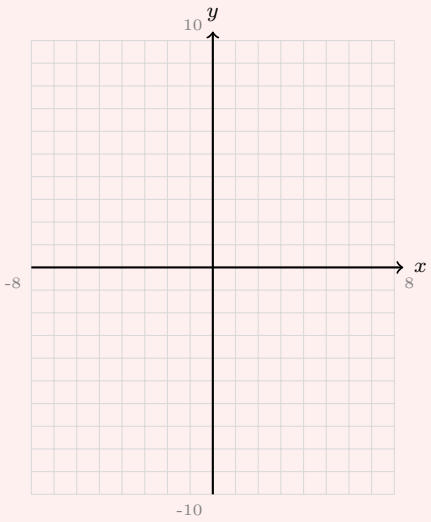
Hard

(a) Graph $y = -x^2 + 2x + 8$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down

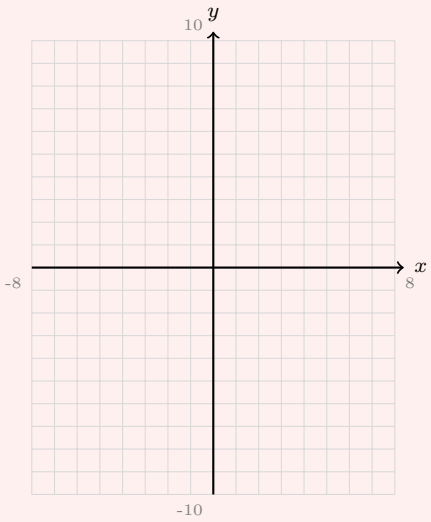


(b) Graph $y = 2x^2 - 8x + 3$ (use the vertex/quadratic formula — roots are irrational).

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down



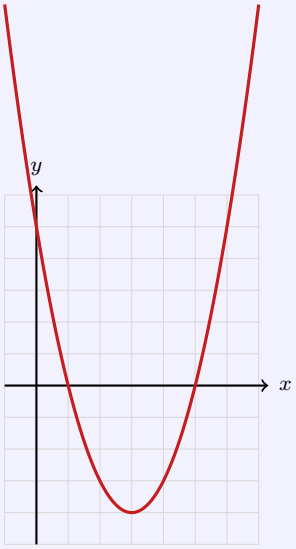
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A.2 Vertex Form: $y = a(x-h)^2 + k$

Worked Example

Graph $y = (x-3)^2 - 4$.

- Vertex: $(h,k) = (3,-4)$ read directly
- Axis of symmetry: $x = 3$
- y-intercept: $x = 0 \Rightarrow y = (0-3)^2 - 4 = 5 \Rightarrow (0,5)$
- x-intercepts: $(x-3)^2 - 4 = 0 \Rightarrow (x-3)^2 = 4 \Rightarrow x - 3 = \pm 2 \Rightarrow x = 1, 5$
- $a = 1 > 0 \Rightarrow$ opens up



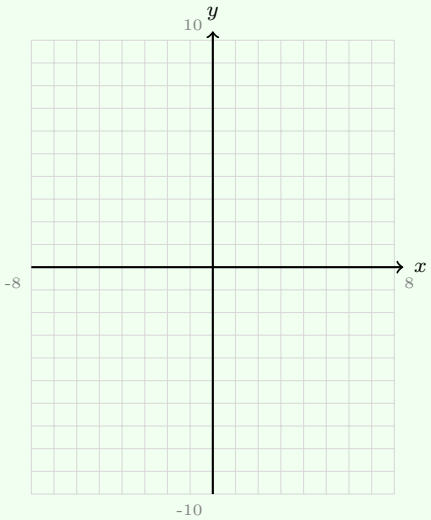
Easy

(a) Graph $y = (x-2)^2 + 1$ (hint: no real x-intercepts).

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down

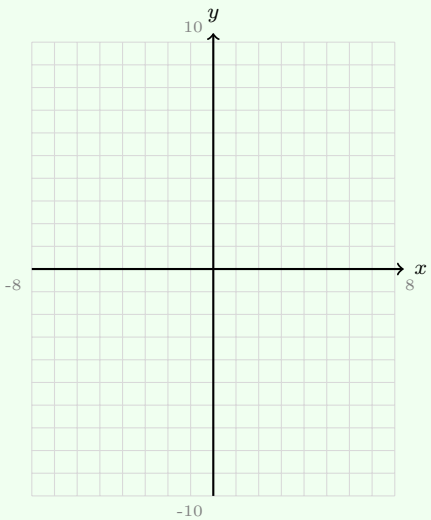


(b) Graph $y = (x+1)^2 - 4$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down



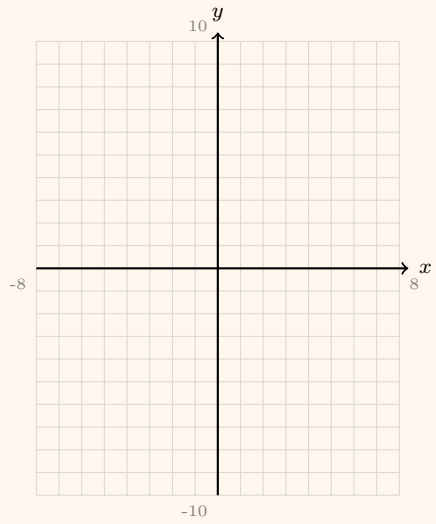
Medium

(a) Graph $y = -2(x-1)^2 + 8$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down

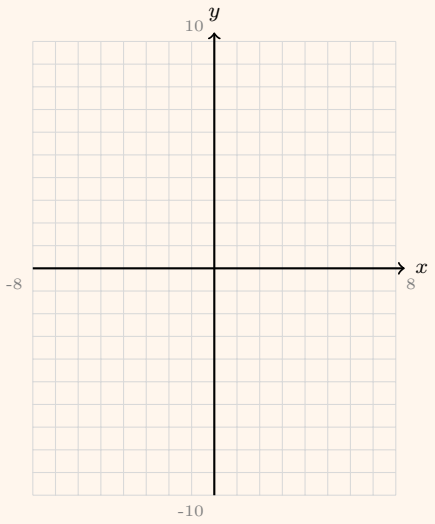


(b) Graph $y = \frac{1}{2}(x+3)^2 - 2$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down



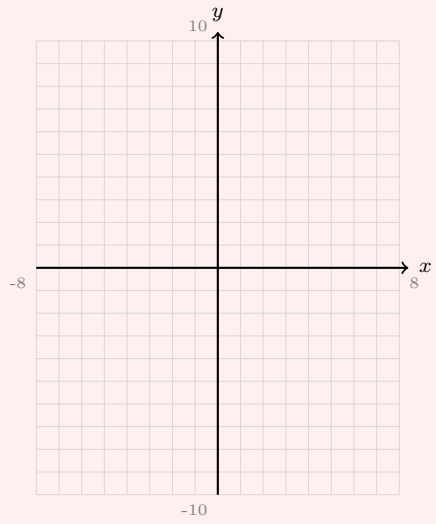
Hard

(a) Graph $y = -3(x+2)^2 + 5$ (x-int. irrational).

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down

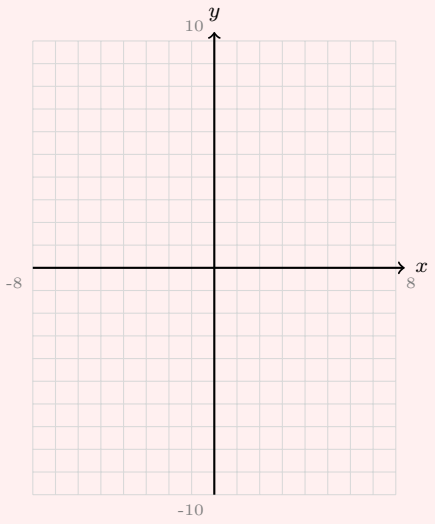


(b) Graph $y = 2(x-1)^2 - 7$ (x-int. irrational).

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down



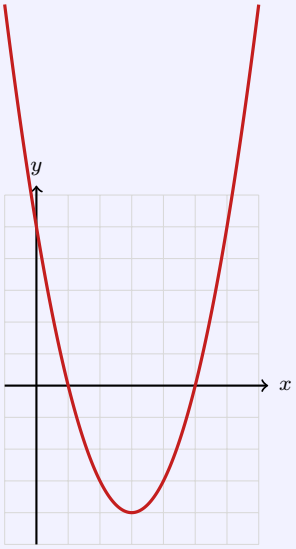
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A.3 Intercept (Factored) Form: $y = a(x-p)(x-q)$

Worked Example

Graph $y = (x-1)(x-5)$.

- x-intercepts: $x = 1$ and $x = 5$ read directly
- Axis of symmetry: $x = \dfrac{1+5}{2} = 3$
- Vertex: $x = 3 \Rightarrow y = (3-1)(3-5) = 2(-2) = -4 \Rightarrow (3, -4)$
- y-intercept: $x = 0 \Rightarrow y = (-1)(-5) = 5 \Rightarrow (0, 5)$
- $a = 1 > 0 \Rightarrow$ opens up



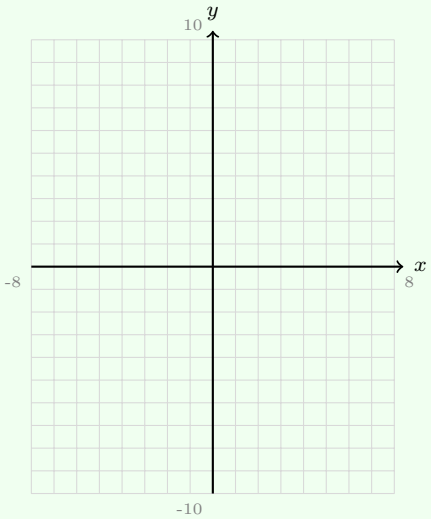
Easy

(a) Graph $y = (x-2)(x-4)$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down

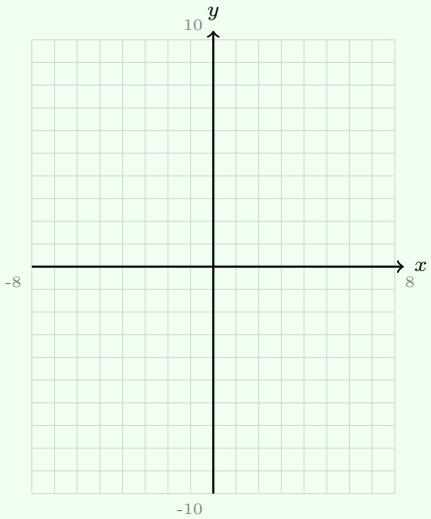


(b) Graph $y = (x+1)(x-3)$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down



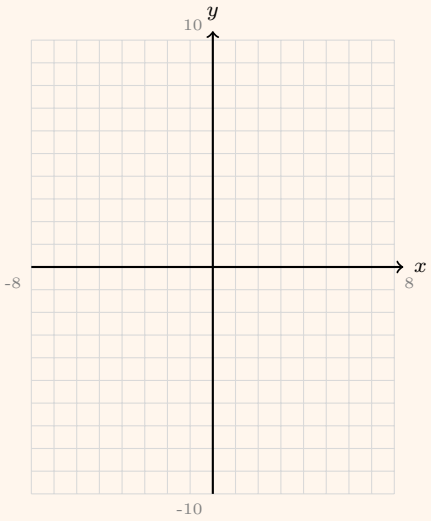
Medium

(a) Graph $y = -(x+2)(x-4)$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down

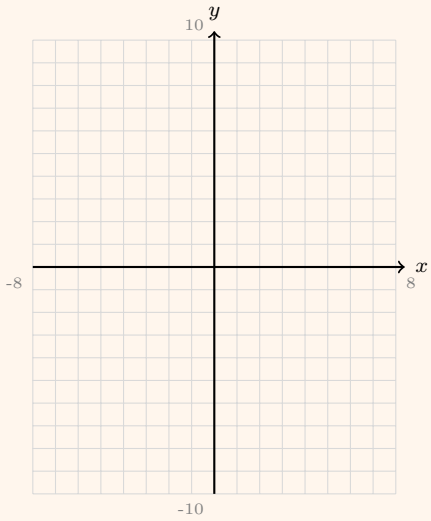


(b) Graph $y = 2(x-1)(x+3)$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down



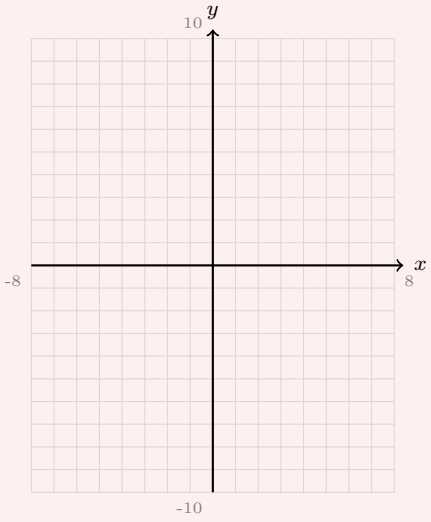
Hard

(a) Graph $y = -2(x+3)(x-1)$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down

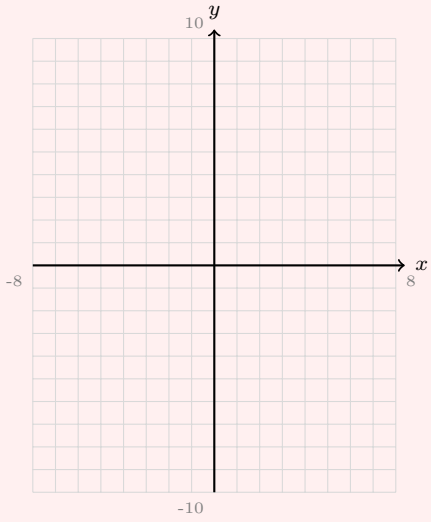


(b) Graph $y = \frac{1}{2}(x-6)(x+2)$.

Vertex: _____ Axis of sym.: _____

x-int.: _____ y-int.: _____

Opens: ◯ Up ◯ Down



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Part B · Exponential Functions — Domain and Range

An exponential function has the form $f(x) = a \cdot b^{x-h} + k$, where $b > 0$, $b \neq 1$. Transformations shift the graph but never change its basic shape.

Domain: always all real numbers, $(-\infty, \infty)$, for every exponential function of this form — there is no restriction on x.

Range: determined by the horizontal asymptote $y = k$ and the sign of a:

- If $a > 0$: range is $y > k$
- If $a < 0$: range is $y < k$

Worked Example

Find the domain and range of $f(x) = 3^x + 2$.

- Domain: all real numbers, $(-\infty, \infty)$
- Horizontal asymptote: $y = 2$; since $a = 1 > 0$, the graph lies above the asymptote
- Range: $y > 2$, i.e. $(2, \infty)$

Hard

Find the domain and range of each function.

(a) $f(x) = -2^x - 3$

(b) $f(x) = 5\left(\frac{1}{2}\right)^{x-1} + 4$

(c) $f(x) = -3 \cdot 2^{x+2} - 1$

Part C · Arithmetic and Geometric Sequences

Arithmetic sequence: each term changes by adding a constant **common difference** d.

$$a_n = a_1 + (n-1)d$$

Geometric sequence: each term changes by multiplying by a constant **common ratio** r.

$$a_n = a_1 \cdot r^{n-1}$$

Worked Example

Arithmetic: Find a_{10} of $5, 9, 13, 17, \ldots$
$a_1 = 5$, $d = 4 \Rightarrow a_{10} = 5 + (10-1)(4) = 5 + 36 = 41$

Geometric: Find a_8 of $3, 6, 12, 24, \ldots$
$a_1 = 3$, $r = 2 \Rightarrow a_8 = 3 \cdot 2^7 = 3(128) = 384$

Easy

(a) Arithmetic: Find a_{12} of the sequence $4, 7, 10, 13, \ldots$

(b) Geometric: Find a_6 of the sequence $2, 6, 18, 54, \ldots$

Medium

(a) Arithmetic: An arithmetic sequence has $a_1 = 6$ and common difference $d = 4$. Find a_{15}.

(b) Geometric: A geometric sequence has $a_1 = 100$ and common ratio $r = 0.5$. Find a_5.

Hard

(a) Arithmetic: In an arithmetic sequence, $a_1 = -5$, $d = 4$, and $a_n = 67$. Find n.

(b) Geometric: In a geometric sequence, $a_1 = 4$, $r = 3$, and $a_n = 2916$. Find n.

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Answer Key

Part A — Quadratic Functions

Standard Form Example: $y = x^2 - 4x + 3$: vertex $(2, -1)$, axis $x = 2$, x-int. $1, 3$, y-int. 3, opens up.

Easy: (a) $y = x^2 - 2x - 3$: vertex $(1, -4)$, axis $x = 1$, x-int. $-1, 3$, y-int. -3, opens up. (b) $y = x^2 + 4x$: vertex $(-2, -4)$, axis $x = -2$, x-int. $0, -4$, y-int. 0, opens up.

Medium: (a) $y = x^2 - 6x + 5$: vertex $(3, -4)$, axis $x = 3$, x-int. $1, 5$, y-int. 5, opens up. (b) $y = 2x^2 + 4x - 6$: vertex $(-1, -8)$, axis $x = -1$, x-int. $-3, 1$, y-int. -6, opens up.

Hard: (a) $y = -x^2 + 2x + 8$: vertex $(1, 9)$, axis $x = 1$, x-int. $4, -2$, y-int. 8, opens down. (b) $y = 2x^2 - 8x + 3$: vertex $(2, -5)$, axis $x = 2$, x-int. $x = 2 \pm \frac{\sqrt{10}}{2} \approx 3.58, 0.42$, y-int. 3, opens up.

Vertex Form Example: $y = (x - 3)^2 - 4$: vertex $(3, -4)$, axis $x = 3$, x-int. $1, 5$, y-int. 5, opens up.

Easy: (a) $y = (x - 2)^2 + 1$: vertex $(2, 1)$, axis $x = 2$, no real x-int., y-int. 5, opens up. (b) $y = (x + 1)^2 - 4$: vertex $(-1, -4)$, axis $x = -1$, x-int. $1, -3$, y-int. -3, opens up.

Medium: (a) $y = -2(x-1)^2+8$: vertex $(1, 8)$, axis $x = 1$, x-int. $3, -1$, y-int. 6, opens down. (b) $y = \frac{1}{2}(x+3)^2-2$: vertex $(-3, -2)$, axis $x = -3$, x-int. $-1, -5$, y-int. 2.5, opens up.

Hard: (a) $y = -3(x + 2)^2 + 5$: vertex $(-2, 5)$, axis $x = -2$, x-int. $x = -2 \pm \sqrt{5/3} \approx -0.71, -3.29$, y-int. -7, opens down. (b) $y = 2(x - 1)^2 - 7$: vertex $(1, -7)$, axis $x = 1$, x-int. $x = 1 \pm \sqrt{3.5} \approx 2.87, -0.87$, y-int. -5, opens up.

Intercept Form Example: $y = (x - 1)(x - 5)$: vertex $(3, -4)$, axis $x = 3$, x-int. $1, 5$, y-int. 5, opens up.

Easy: (a) $y = (x - 2)(x - 4)$: vertex $(3, -1)$, axis $x = 3$, x-int. $2, 4$, y-int. 8, opens up. (b) $y = (x + 1)(x - 3)$: vertex $(1, -4)$, axis $x = 1$, x-int. $-1, 3$, y-int. -3, opens up.

Medium: (a) $y = -(x + 2)(x - 4)$: vertex $(1, 9)$, axis $x = 1$, x-int. $-2, 4$, y-int. 8, opens down. (b) $y = 2(x - 1)(x + 3)$: vertex $(-1, -8)$, axis $x = -1$, x-int. $1, -3$, y-int. -6, opens up.

Hard: (a) $y = -2(x + 3)(x - 1)$: vertex $(-1, 8)$, axis $x = -1$, x-int. $-3, 1$, y-int. 6, opens down. (b) $y = \frac{1}{2}(x - 6)(x + 2)$: vertex $(2, -8)$, axis $x = 2$, x-int. $6, -2$, y-int. -6, opens up.

Part B — Exponential Domain & Range

Example: $f(x) = 3^x + 2$: domain $(-\infty, \infty)$, range $y > 2$.

(a) $f(x) = -2^x - 3$: domain $(-\infty, \infty)$; asymptote $y = -3$, $a = -1 < 0$, range $y < -3$.
(b) $f(x) = 5(\frac{1}{2})^{x-1} + 4$: domain $(-\infty, \infty)$; asymptote $y = 4$, $a = 5 > 0$, range $y > 4$.
(c) $f(x) = -3 \cdot 2^{x+2} - 1$: domain $(-\infty, \infty)$; asymptote $y = -1$, $a = -3 < 0$, range $y < -1$.

Part C — Arithmetic & Geometric Sequences

Example: Arithmetic $a_{10} = 41$. Geometric $a_8 = 384$.

Easy: (a) $a_{12} = 4 + (11)(3) = 37$. (b) $a_6 = 2 \cdot 3^5 = 486$.
Medium: (a) $a_{15} = 6 + (14)(4) = 62$. (b) $a_5 = 100(0.5)^4 = 6.25$.
Hard: (a) $67 = -5 + (n - 1)(4) \Rightarrow n = 19$. (b) $2916 = 4 \cdot 3^{n-1} \Rightarrow 3^{n-1} = 729 = 3^6 \Rightarrow n = 7$.

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